School of Mechanical and Manufacturing Engineering
Faculty of Engineering
UNSW Sydney

**Aidan Knipe**
z5418749

# Macro-Micro Robotic System with a Flexible Continuum Tip for Minimally Invasive Surgery

Interim Report and Project Plan

**Abstract**

This interim report and project plan is a Thesis A submission that lays the groundwork for improvements to a UR5e-based macro-micro robotic system with a flexible continuum tip for minimally invasive surgery (MIS). It consolidates relevant findings from prior work and available literature in relevant fields, and critically analyses how these prior works are related to this thesis. The review explores three technical threads relating to improvements of the UR5e-based system. Control improvemments for teleoperation, tool exchange mechanisms for modularity, and safety and reliability considerations are reviewed, with a focus on identifying specific gaps that this thesis can address. The report finds clear research opportunities in these areas, specifically around smoothing for teleoperative control, integrated tool exchange validation, and comprehensive safety analysis.

# Contents


**List of Figures** iv

**List of Tables** v

**Nomenclature** vi

**1 Introduction** 1

1.1 Project Context and Problem Significance . . . . 1

**2 Literature Review** 1

2.1 Minimally Invasive Surgery (MIS) and Robotic Systems . . . . 1

2.1.1 Historical Progression of Surgery and MIS Robotics . . . . 1

2.1.2 Macro–Micro Architecture for Surgical Robotics . . . . 2

2.1.3 Continuum Manipulators for MIS . . . . 3

2.1.4 Cost and Accessibility Considerations . . . . 4

2.1.5 UR5e-Based Systems and Identified Limitations . . . . 4

2.2 Teleoperation and Control in Surgical Robotics . . . . 5

2.2.1 Modelling Frameworks for Continuum Surgical Robotics . . . . 5

2.2.2 Tremor Suppression and Delay-Robust Interaction . . . . 6

2.2.3 State Estimation and Predictive Bilateral Architectures . . . . 7

2.2.4 Shared Control, Immersion, and Haptic Force Awareness . . . . 8

2.2.5 Transmission Nonlinearity and Distal Force Safety . . . . 9

2.3 Modularity and Tendon-Driven Continuum Manipulators . . . . 10

2.3.1 Monolithic vs Modular Distal Architectures . . . . 10

2.3.2 Tool Exchange Mechanisms . . . . 11

2.3.3 Rail-Guided and Cartridge-Based Exchange . . . . 11

2.3.4 Reconfigurable and Multifunctional Instruments . . . . 11

2.3.5 Limitations of Alternative Exchange Mechanisms . . . . 11

2.3.6 Post-Swap Control Recovery and Tension Management . . . . 12

2.3.7 Critical Assessment and Identified Knowledge Gap . . . . 13

2.4 System Safety and Reliability Considerations . . . . 14

2.4.1 Safety Constraint Model and Hazard Taxonomy . . . . 14

2.4.2 Error Detection and Reporting Architecture . . . . 14

2.4.3 Malfunction Frequency and Immediate Clinical Consequence . . . . 15

2.4.4 Malfunction Frequency and Clinical Consequence . . . . 15

2.4.5 Preventable Versus Non-Preventable Faults . . . . 15

2.4.6 Lifecycle Reliability: Reconfiguration, Wear, and Reprocessing . . . . 15

2.4.7 Integrated Safety Gap and Evaluation Endpoints . . . . 16

2.5 Synthesis of Literature and Knowledge Gaps . . . . 16

2.6 Priority Evidence Gaps for Additional Review . . . . . . . . . . . . . . . . . . 17

**3 Research Question and Project Plan** **18**

3.1 Research Question and Plan Status . . . . . . . . . . . . . . . . . . . . . . . . . 18

3.2 Plan Quality Criteria (Assessment-Aligned) . . . . . . . . . . . . . . . . . . . 18

3.3 Current Planning Position . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 18

**4 Project Dependent Preparations** **18**

**5 Conclusion** **18**

**Acknowledgements** **A**

**References** **B**

**Appendices** **E**

**A Rubric-Aligned Coverage Notes (MMAN4951/MMAN9451 Interim Report)** **E**

# List of Figures

1 Progression of surgery and enabling technology from conventional approaches to modern MIS robotics. . . . . . . . . . . . . . . . . . . . . . . . . 2
2 Macro–micro teleoperation system combining a UR5e robotic arm with a continuum-tip manipulator [1]. . . . . . . . . . . . . . . . . . . . . . . . . . . 2
3 System architecture of a UR5e-based macro–micro surgical manipulator, including teleoperation and continuum-tip integration [1]. . . . . . . . . . . . 3
4 SnakeRaven macro–micro surgical manipulator concept [8]. . . . . . . . . . . 4
5 Linked control layers relevant to teleoperation in macro–micro surgical robotics. The literature reviewed in this subsection can be interpreted across these interacting layers, from low-level transmission effects to high-level shared control. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 6
6 Simplified teleoperation architecture for macro–micro surgical robotics. The reviewed literature addresses different parts of this loop, including state estimation, shared control, predictive control, and transmission compensation. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 10

# List of Tables

1 Comparison of tool-exchange strategy families relevant to modular tendon-driven surgical manipulators . . . . . . . . . . . . . . . . . . . . . . . . . . . . 13

# Nomenclature

*(The nomenclature must respect the following order: latin symbols, Greek symbols, acronyms. Within each category of symbols, follow alphabetic order. An example is provided below)*

| | | |
|---|---|---|
| $A_c$ | = | tendon/cable cross-sectional area |
| $D$ | = | backlash deadband |
| $E$ | = | Young's modulus of tendon/cable material |
| $M_{grasp}$ | = | estimated distal grasping torque |
| $R_0$ | = | driving pulley radius |
| $R_5$ | = | driven pulley radius |
| $R_{eq}$ | = | equivalent-circle radius for tendon–sheath geometry |
| $R_i$ | = | orientation matrix of segment $i$ |
| $T_i$ | = | homogeneous transform of segment $i$ |
| $\Delta\theta_{out}$ | = | output-side angular backlash/displacement loss |
| $\Omega$ | = | total cumulated curve angle of a tendon–sheath path |
| $\theta_{comp}$ | = | feedforward backlash compensation term |
| $\mu$ | = | friction coefficient (cable–pulley contact) |
| $\theta_i$ | = | local bending angle of segment $i$ |
| $\alpha_i$ | = | local orientation offset of segment $i$ |
| $\boldsymbol{p}$ | = | rod centerline position in Cosserat formulation |
| $\boldsymbol{R}$ | = | rod orientation in Cosserat formulation |
| $\boldsymbol{n}$ | = | internal force in Cosserat formulation |
| $\boldsymbol{m}$ | = | internal moment in Cosserat formulation |
| MIS | = | minimally invasive surgery |
| TLA | = | ... |

# 1 Introduction

## 1.1 Project Context and Problem Significance

This thesis aims to extend on previous macro-micro robotic surgery work by 2023 UNSW thesis student Lachlan Scott [1]. Scott partially identified areas of improvement that he saw for his own system, which this thesis aims to address. Furthermore, new improvement and research possibilities have been identified that also build on Scott's work. The main areas of possible improvement are in the teleoperation control and motion of the system, the modularity of the system and thus ease of tool exchange, and the safety and reliability of the system for clinical use. Scott's earlier project has established the viability of the system that will be discussed at a proof-of-concept level. Hence, this review of the literature and the thesis question that follow need to recognise this context with regards to Scott's work.

# 2 Literature Review

## 2.1 Minimally Invasive Surgery (MIS) and Robotic Systems

### 2.1.1 Historical Progression of Surgery and MIS Robotics

Technology and surgery have been inseparable for centuries as every surgical advancement or care improvement has been enabled by the parallel development of instruments and tools. From anaesthesia to antiseptics, innovation has enabled surgeons to reduce risk and expand the scope of treatable conditions [2]. In the 19th century, innovations such as the transition from candlelight to electric lighting allowed for practical internal visualisation, marking the beginning of a major paradigm shift in the surgical field (Hargest five thousand years). Electricity also enable the development of various specialised surgical procedures, such as electrocautery for haemostatasis and the use of electric drills for orthopaedic surgery [3]. These innovations allowed for a stronger focus on precision and dexterity, and techniques such as laparoscopy and endoscopy were developed at the beginning of the 20th century, resulting in some of the first minimally invasive surgeries (MIS) [4]. This trend towards less invasive procedures has continued, with the development of revolutionary robotic surgical systems such as the da Vinci Surgical System in the late 1990s, which has become widely adopted for various surgical procedures [5]. Laparoscopic surgery is a core MIS approach in which instruments are inserted through small abdominal incisions using trocar access ports, rather than a large open incision [6]. In this report, MIS denotes procedures that use these reduced-access pathways to lower tissue trauma and recovery burden relative to open surgery.

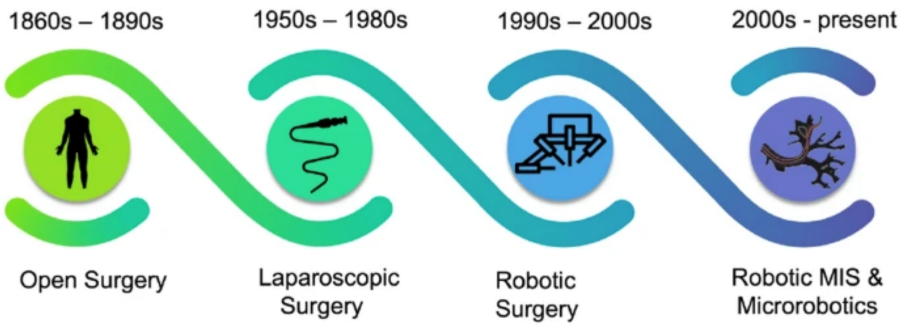



Figure 1: Progression of surgery and enabling technology from conventional approaches to modern MIS robotics.

#### 2.1.2 Macro–Micro Architecture for Surgical Robotics

The current aspiration of modern surgical robotics is achieving human-level dexterity, accuracy, and force control in increasingly small and complex anatomical spaces that may have limited access or visualisation. One key technological direction enabling this transition is minimally invasive surgical tooling with a macro–micro architecture. This terminology refers to a two-stage robotic system in which a larger, more rigid macro manipulator provides gross positioning and stability, while a smaller, more flexible micro manipulator is attached to the distal end of the macro arm to provide fine manipulation capabilities in confined spaces [1].

The macro unit provides large-workspace positioning and efficiency; however, its larger structure can be prone to deformation and vibration. In contrast, the micro manipulation unit provides the distal dexterity required for manipulation in confined anatomy, where surgical targets often require end-effectors in the 3–4 mm class for practical access through narrow channels and natural orifices.

An example of this macro–micro architecture is shown in Fig. 2, where a UR5e macro robotic arm is integrated with a tendon-driven continuum micro-manipulator [1].

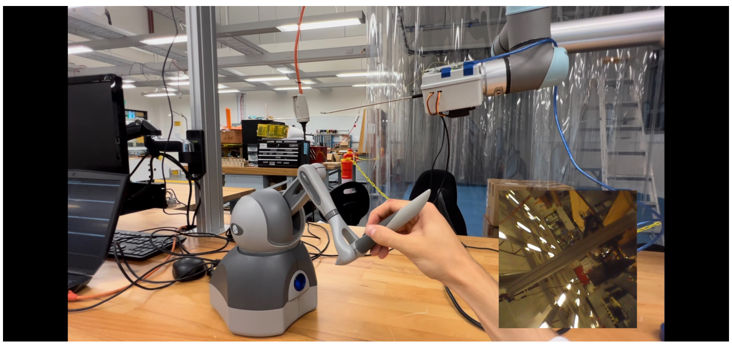


Figure 2: Macro–micro teleoperation system combining a UR5e robotic arm with a continuum-tip manipulator [1].

In current snake-robot literature, this micro stage is typically tendon or cable-actuated to achieve high local curvature and fine orientation control, but its performance depends strongly on motion transmission quality, sensing, and control architecture [7]. Within a macro–micro arrangement, this local dexterity complements the macro arm's gross positioning capability and can help isolate the endpoint from macro-stage vibration through coordinated control [1].

A system-level representation of this architecture is shown in Fig. 3, illuminating the component integration.

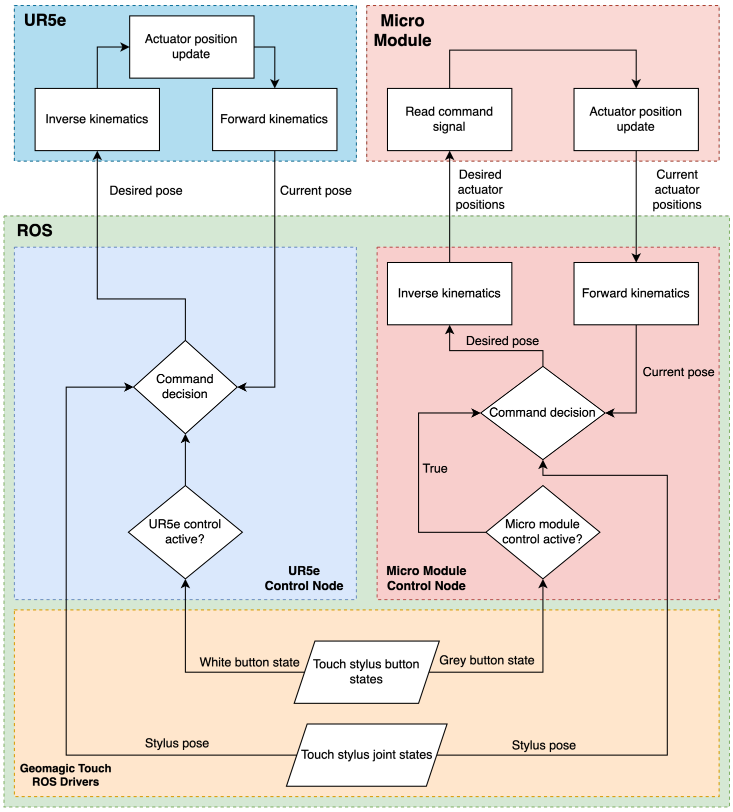
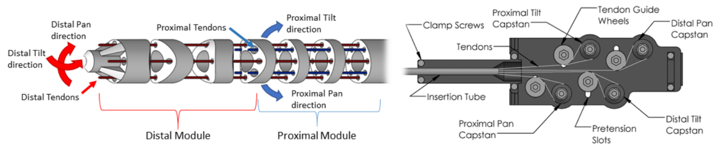



Figure 3: System architecture of a UR5e-based macro–micro surgical manipulator, including teleoperation and continuum-tip integration [1].

### 2.1.3 Continuum Manipulators for MIS

This review is primarily focused on continuum tips, which are a type of hyper-redundant end-effectors for MIS. They combine high dexterity with slender geometries so that they can inserted through tightly constrained pathways (e.g., lumens, natural orifices, and smallincisions), where traditional rigid tools may not fit.

In the tendon-driven subclass, curvature is generated through differential tendon tension about a central backbone or articulated serial structure, enabling multi-DOF distal steering with compact proximal actuation and making these devices well suited to macro-micro architectures [7]. An example of this is shown in Fig. 4, where a tendon-driven continuum tip is integrated with a Raven II surgical robot for teleoperation [8].




**(a)** Diagram of the Two-module End-effector **(b)** Interior of the adaptor to the RAVEN II

Figure 4: SnakeRaven macro–micro surgical manipulator concept [8].

#### 2.1.4 Cost and Accessibility Considerations

With growing demand and a rapidly expanding surgical robotics market, there is ongoing debate over whether the high cost of integrated platforms such as the da Vinci system limits accessibility. Acquisition costs have been reported up to

$4 million AUD, with additional per-procedure costs of approximately $1000–$5000 [9].

Despite clear clinical benefits, providers must still evaluate cost-effectiveness, particularly in resource-limited settings [9]. Evidence from low and middle-income environments shows that there are more uptake barriers than the initial cost. Recurring maintenance, disposable instrument costs, and sustained training requirements can prevent programme continuity even after installation [10].

At a system level, further constraints arise from limited infrastructure, workforce training gaps, and the need to prioritise basic surgical capacity over advanced robotic systems [10]. These factors collectively highlight a significant translational gap between technological capability and real-world clinical deployment.

Taken together, the literature indicates that scalable, cost-effective robotic solutions remain underdeveloped. In particular, there is a need for affordable macro–micro systems that combine low-cost hardware with clinically credible and robust teleoperation performance.

This aligns with broader research directions advocating the use of off-the-shelf hardware and open-source software frameworks to reduce system cost while maintaining the precision and safety requirements of minimally invasive surgery [1].

#### 2.1.5 UR5e-Based Systems and Identified Limitations

The UR5e is a widely used, cost-effective robotic manipulator that can serve as the macro stage in macro–micro surgical architectures, with reported pricing around $50,000 to $60,000 AUD [11]. Its relatively low cost and flexibility make it a suitable platform for integration with custom micro-manipulation systems.

The feasibility of a UR5e-based macro–micro architecture has been demonstrated in prior thesis work by Scott (2022), which implemented a proof-of-concept teleoperation pipeline using open-source ROS2. Within this framework, a tendon-driven continuum micro-manipulator was integrated with the macro arm and operated via a haptic interface, following the snake-like design approaches commonly reported in the literature.

Preliminary validation using electromagnetic tracking demonstrated close pose mirroring under teleoperation, with reported average absolute position errors of 0.0272 mm and 0.0161 mm across trial groups. These results indicate that high-precision control is achievable within such macro–micro configurations.

However, several practical engineering limitations remain. In particular, translational commands at the macro stage were observed to induce vibration in the integrated system, negatively affecting motion stability and repeatability. Additionally, the current modular workflow requires manual exchange of tendon-driven end-effectors, resulting in increased setup time and reduced operational efficiency.

These limitations highlight a key gap in existing UR5e-based macro–micro systems: while feasibility and precision have been demonstrated, improvements in vibration suppression, system robustness, and workflow integration are required to meet clinical expectations for safety and usability in real-world operating environments [1], [7], [8].

## 2.2 Teleoperation and Control in Surgical Robotics

### 2.2.1 Modelling Frameworks for Continuum Surgical Robotics

To situate control and modularity claims on a consistent mathematical basis, three modelling layers from continuum-robotics literature are directly relevant to this project: kinematic representations, reduced-order curvature models, and mechanics-based formulations. At the kinematic level, each segment transform can be expressed as

$$T_i = \begin{bmatrix} R_i & p_i \; \mathbf{0}^\top & 1 \end{bmatrix}, \tag{1}$$

which maps segment states to distal pose and forms the baseline representation used across continuum and hyper-redundant robotic systems [12], [13]. This formulation is computationally efficient and well suited to real-time teleoperation, but does not explicitly capture load-dependent deformation.

For tendon-driven continuum manipulators, piecewise-constant-curvature (PCC) models provide a practical actuator-to-workspace mapping:

$$T_i^{i+1} = \mathrm{rot}(Z_i, \alpha_i), \mathrm{trans}(Z_i, H_i), \mathrm{trans}(X_i, W_i), \mathrm{rot}(Y_i, \theta_i), \mathrm{rot}(Z_i, -\alpha_i), \tag{2}$$

which underpins many high-DOF modelling and calibration approaches [13], [14]. PCC models strike a balance between computational efficiency and geometric fidelity, making them widely adopted in teleoperated continuum systems.

When compliance, external loading, and routing-dependent behaviour become dominant, mechanics-based approaches such as Cosserat-rod models provide a more complete representation of continuum dynamics. These models capture distributed force and moment balance along the manipulator backbone, enabling analysis of tendon friction, hysteresis, and deformation under load **janabi2021continuum**.

In critical terms, the literature indicates that no single modelling layer is sufficient for workflow-ready surgical performance: kinematic models are fast but approximate, while full mechanics-based models are more expressive but computationally intensive and harder to identify in real time. This trade-off directly motivates hybrid modelling and compensation strategies in modern teleoperation systems.

Within this control context, the relevant evidence can be interpreted across five linked layers: tremor suppression, latency compensation, robust state estimation, predictive bilateral control, and intent-aware shared autonomy. This layered interpretation is summarised in Fig. 5.

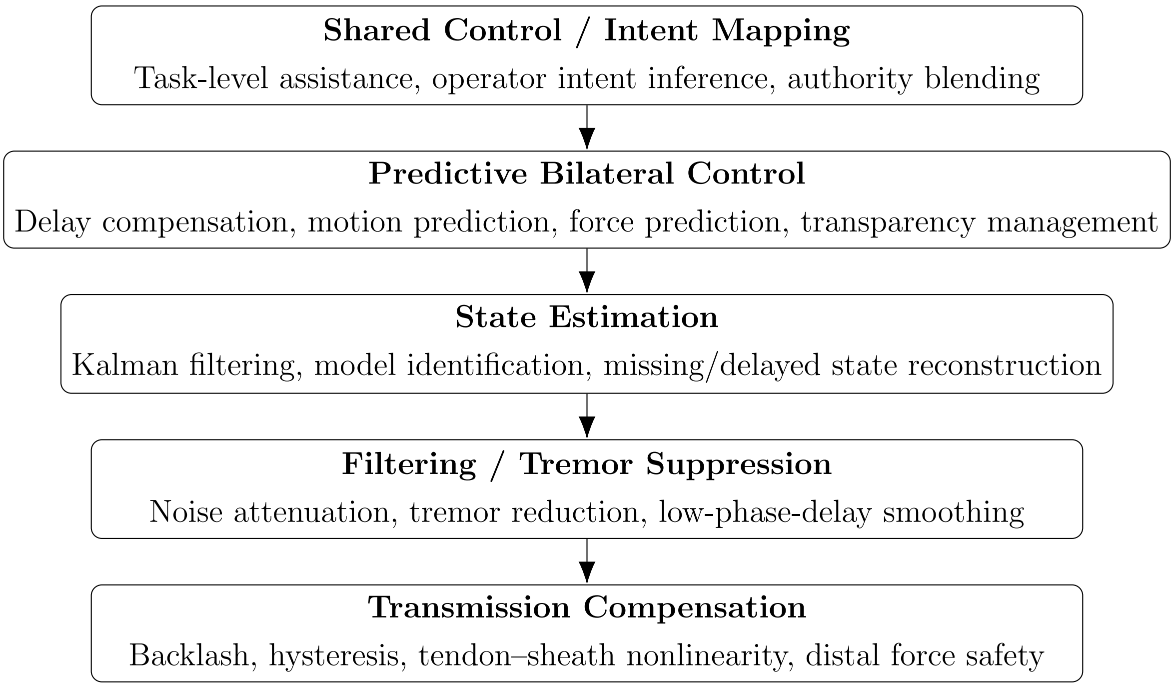



Figure 5: Linked control layers relevant to teleoperation in macro–micro surgical robotics. The literature reviewed in this subsection can be interpreted across these interacting layers, from low-level transmission effects to high-level shared control.

### 2.2.2 Tremor Suppression and Delay-Robust Interaction

Human physiological tremor is a central control challenge in robot-assisted MIS because involuntary operator motion is transmitted through the teleoperation loop and can appear as vibration at the surgical instrument tip, degrading precision during fine tissue manipulation, suturing, and needle handling. Sang, Yang, Liu, *et al.* [15] report that tremor suppression methods must balance two competing requirements: strong attenuation of involuntary motion and minimal phase delay, as latency in the command pathway can directly reduce teleoperation quality. Their zero-phase adaptive fuzzy Kalman filter (ZPAFKF) combines a zero-phase prefilter with adaptive Kalman estimation to improve tremor estimation and compensation in real time, and experimental master–slave results show lower error and reduced tip vibration relative to standard filtering baselines. This is directly relevant to the present thesis because it frames tremor control not as a standalone signal-processing problem, but as a coupled teleoperation and tracking problem in which compensation quality, delay, and manipulator dynamics must be considered together [15].

System latency is a second critical bottleneck in bilateral teleoperation, particularly when haptic feedback is included. Batty, Ehrampoosh, Shirinzadeh, *et al.* [16] show that communication delay creates a fundamental transparency–stability trade-off: increasing direct force correspondence can degrade stability, while aggressive stabilisation can mute the fidelity of operator feedback. In response, they implement an environment-estimation and force-prediction architecture, where slave-side contact parameters are estimated online (EWRLS) and master feedback is generated through a predicted virtual environment rather than delayed direct force reflection. Their human-in-the-loop palpation experiments demonstrate that this predictive approach substantially reduces force prediction error (with the Hunt–Crossley model outperforming Kelvin–Voigt and direct force feedback), and improves teleoperation transparency under delay. A key implication in the literature is that latency management should be treated as a control-architecture problem, not solely a communications problem, and that model quality strongly influences the practical quality of teleoperated manipulation [16]. Effective solutions must therefore balance filtering, prediction, and system dynamics to maintain both stability and responsiveness.

### 2.2.3 State Estimation and Predictive Bilateral Architectures

Filtering is therefore not optional in surgical teleoperation; it is a foundational requirement for maintaining stable and trustworthy state estimates when measurements are noisy and network transport is imperfect. Lashari, Batayneh, and Khokhar [17] show that a Kalman filter paired with MOESP-identified state-space models can estimate patient-side manipulator position with high robustness under delay, jitter, and packet loss, while remaining computationally lightweight for real-time use. This is especially relevant when exact plant dynamics are not available a priori, because MOESP builds the model directly from surgical kinematic data (JIGSAWS), allowing filtering performance to be grounded in observed system behaviour rather than idealised assumptions. In practical terms, filters are needed to smooth stochastic disturbances, reconstruct missing or delayed information, and preserve control continuity so that downstream motion control and haptic/visual feedback remain clinically usable. A key limitation, however, is that the reported performance is primarily based on dataset-driven modelling and simulated network impairments, so in vivo generalisability and cross-platform transfer still require further validation [17].

Predictive bilateral control is also emerging as a practical strategy for continuum and snake-like teleoperation under delay. Ebrahimian, Pourmokhtari, Ghiyasi, *et al.* [18] propose an online architecture that combines model-mediated force prediction on the master side with ANN-based operator motion prediction on the slave side, with the explicit goal of reducing perceived communication lag while preserving position synchronisation and haptic transparency. This is highly relevant to the macro–micro literature because it addresses the same core bottleneck as latency and filtering studies, but at the controller-architecture level: delay compensation is handled through coordinated forward and backward prediction, rather than through one-sided compensation only. Their simulation results show strong delay bypass for known constant delays in snake-like robot teleoperation, but a key limitation is that performance is demonstrated mainly in simulation with structured assumptions (e.g., known constant delay and constrained robot DOFs), which may not fully capture variable-delay clinical networks or higher-complexity surgical interaction dynamics [18]. Together, these approaches highlight that robust teleoperation under delay requires both reliable state estimation and predictive control, reinforcing the need for integrated architectures rather than isolated solutions.

#### 2.2.4 Shared Control, Immersion, and Haptic Force Awareness

Beyond tremor and delay, user intent mapping is a central teleoperation design problem in shared control. Bowman, Zhang, and Zhang [19] argue that direct motion mapping alone is insufficient for robust telemanipulation when the human and robot hands are physically dissimilar, because low-level geometric tracking does not necessarily preserve high-level task success. Their intent-based task-oriented shared control framework instead blends operator-following behaviour with autonomous task assistance using inferred intent probabilities and adaptive arbitration weights. Critically, this contribution is not just another blending policy: it formalises how authority should shift based on hand-structure discrepancy and task constraints, which is directly relevant to macro–micro MIS systems where distal tools cannot perfectly mimic surgeon hand kinematics. This supports the broader literature position that intuitive teleoperation should be evaluated by both task completion and perceived control, rather than endpoint tracking error alone [19].

However, an important limitation is that the reported validation is constrained to controlled grasping tasks and specific intent models, so transferability to highly dynamic, deformable, and safety-critical in vivo surgical contexts remains an open question.

Recent review-level synthesis also indicates that immersive visualisation and haptic augmentation are transitioning from laboratory prototypes toward clinically relevant teleoperation workflows [20]. Across the studies summarised, VR-based viewpoint control and stereoscopic scene rendering improve situational awareness in tasks where fixed-camera teleoperation typically underperforms. Representative evidence includes an increase in pick-and-place success from 62.5% to 90.0% in a viewpoint-independent VR interface trial, and reduced operator workload and task time in mixed-reality assisted manipulation experiments [20].

For MIS relevance, the strongest transferable point is not immersion alone but the interaction between perception quality and force regulation. Bogue [20] report that force-feedback-enabled surgical platforms are now appearing in commercial systems, including preclinical da Vinci 5 results where surgeons applied 43% less tissue force with haptic-enabled instruments, and Saroa trial results showing mean grasp-force reduction from 2.14 N to 0.63 N in a delicate transfer task. These results support including haptic and force-awareness as a safety-linked teleoperation design variable in this thesis, rather than treating it as an optional interface enhancement.

The key limitation is evidentiary heterogeneity: the article aggregates diverse platforms, task types, and evaluation protocols, so direct cross-study performance comparison is limited. For thesis use, this source is therefore most valuable as trend evidence motivating evaluation criteria such as situational awareness, workload, and applied-force moderation, while primary validation claims should still rely on the underlying experimental studies.

### 2.2.5 Transmission Nonlinearity and Distal Force Safety

Transmission backlash in cable-pulley drives is another directly relevant control bottleneck for laparoscopic end-effectors because direction reversals create delay and transition phases in which distal motion and torque do not track proximal commands linearly. Zou, Li, and Huang [21] model this explicitly and use a feedforward compensation term linked to output-side backlash, $\theta_{comp} = \Delta\theta_{out}, R_5/R_0$, together with grasping torque estimation and saturation limiting, $M_{grasp} = M_{out} - M_{fric}$, to reduce tracking error and limit unsafe distal force. Their results indicate meaningful improvements in position and force control for cable-pulley laparoscopic instruments: reported distal-grasper position error is reduced to approximately 1.03° mean (3.49° max), and grasping torque tracking error is reported below 0.025 N·m in compensated instrument tests. Nonetheless, residual error around direction reversal remains, highlighting that compensation quality still depends on parameter identification and friction-model fidelity [21].

Related tendon–sheath evidence further strengthens this point in shape-variable conditions. Hong, Hong, Kim, *et al.* [22] model backlash-like hysteresis in tendon–sheath pairs using an equivalent-circle representation, where deadband is linked to master/slave deformation mismatch ($D = \delta_{TSM(master)} - \delta_{TSM(slave)}$) and then compensated by feedforward updated from proximal calibration without distal sensing. Their results show RMSE reduction from 1.31 mm to 0.19 mm on average (85.5reduction). This is highly relevant to laparoscopic routing uncertainty; however, the approach still depends on reliable calibration and assumptions about unchanged geometric conditions during each compensation interval [22].

Collectively, these studies highlight that transmission nonlinearity remains a critical limitation in tendon-driven surgical manipulators, particularly under variable routing and loading conditions. While feedforward compensation can significantly reduce tracking and force errors, performance remains dependent on accurate parameter identification and stable system geometry.

These findings establish clear performance requirements for low-cost macro–micro platforms and provide a direct conceptual bridge to modular continuum hardware discussions. The relationship between these control problems within the overall teleoperation loop is illustrated in Fig. 6.

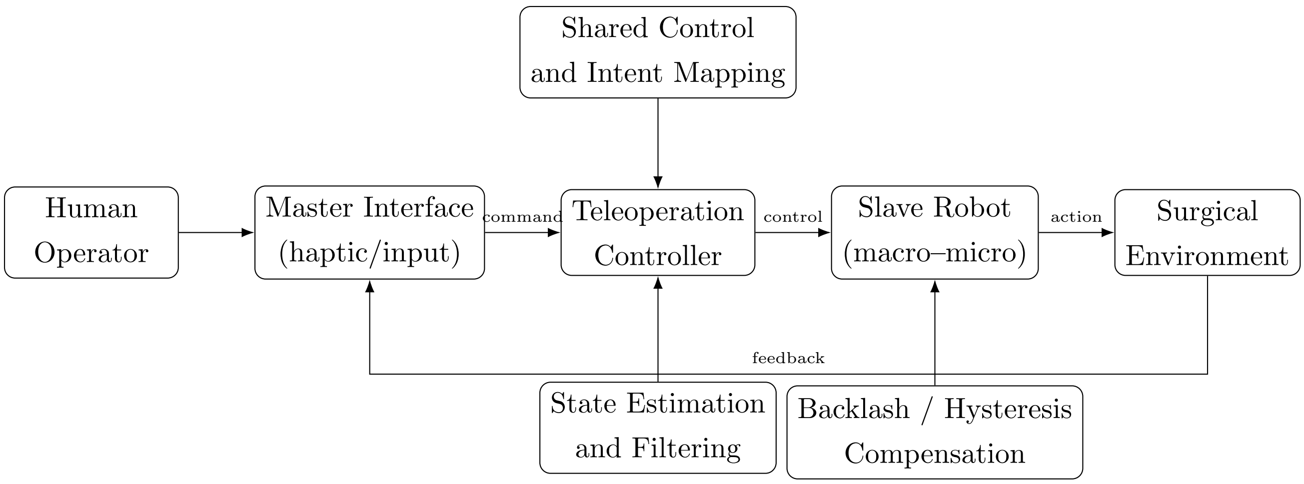



Figure 6: Simplified teleoperation architecture for macro–micro surgical robotics. The reviewed literature addresses different parts of this loop, including state estimation, shared control, predictive control, and transmission compensation.

## 2.3 Modularity and Tendon-Driven Continuum Manipulators

### 2.3.1 Monolithic vs Modular Distal Architectures

In this review, a *monolithic* distal architecture is defined as a single integrated tool assembly in which changing function requires major teardown or full distal module replacement, rather than rapid local exchange. This framing is consistent with the architecture-level distinction between integrated and modular robotic platforms reported by Cowan, Gomes, Morris, *et al.* [23].

The practical consequences are workflow-critical, including longer swap times, re-tensioning and re-calibration burden after replacement, interruption of the operative sequence, and reduced intraoperative adaptability when task requirements change mid-procedure. These constraints align with broader integration-friction evidence in modular end-effector software and hardware pipelines [24], where even technically feasible modularity often carries non-trivial changeover overhead.

Internal platform observations in prior local UR5e–continuum implementations also indicate that end-effector replacement is time-consuming and introduces practical workflow friction in achieving repeatable setup after each swap [1]. Together, the literature and local prior-work evidence indicate that architectural modularity alone is insufficient unless exchange time and post-swap recalibration costs are explicitly addressed.

### 2.3.2 Tool Exchange Mechanisms

Tool exchange is necessary because laparoscopic workflows require multiple functional instruments within a single procedure, while only a limited subset can be mounted simultaneously on the robotic arm [6]. This creates a recurrent intraoperative transition task that can interrupt surgical flow and, when assistant-dependent, introduces operator–assistant coordination variability.

### 2.3.3 Rail-Guided and Cartridge-Based Exchange

At present, the strongest direct evidence remains with rail-guided, cartridge-like exchange approaches. Ko, Kim, Lim, *et al.* [6] analyse current sliding exchange practice and report an automated replacement architecture designed to preserve trocar and remote centre of motion (RCM) constraints via linear motion. Their prototype-level results report full exchange in 15 s, reduced from an initial 27 s cycle, and a total device mass of 3.36 kg (below their 4 kg design target), indicating feasible automation under laparoscopic geometric constraints.

### 2.3.4 Reconfigurable and Multifunctional Instruments

Alternative strategies aim to reduce exchange frequency rather than improve exchange interfaces directly. Mocellin, Pagliarani, Gamberini, *et al.* [25] present OriGrasp as a reconfigurable multifunctional end-effector (introducer mode and compliant grasper mode), validated through dVRK integration and tissue-interaction tests. Reported performance includes approximately 4 N pinch force and bowel pulling/lifting forces in literature-consistent atraumatic ranges. This supports multifunctionality as a useful partial mitigation for selected abdominal tasks. However, current evidence remains task-specific and does not demonstrate that a single reconfigurable device can replace the broader instrument set required across heterogeneous procedures.

### 2.3.5 Limitations of Alternative Exchange Mechanisms

Mechanism diversity beyond rail-guided exchange remains under-validated in the reviewed evidence base. Direct surgical evidence is limited for bayonet quarter-turn, cam-latch, and cartridge-head variants beyond linear sliding, as well as kinematic-coupling-assisted docking with active preload. Reported gaps are not purely mechanical: tolerance sensitivity, sterilisation robustness, wear-dependent repeatability drift, and post-reconnection re-zeroing burden all remain insufficiently quantified across repeated swap cycles.

#### 2.3.6 Post-Swap Control Recovery and Tension Management

Control evidence indicates that docking success alone is insufficient for workflow-ready exchange. Cable-pulley and tendon–sheath studies report backlash, hysteresis, and force-estimation sensitivity to pretension, friction, and routing geometry [21], [22]. In particular, Hong, Hong, Kim, *et al.* [22] show geometry-dependent deadband behaviour and report RMSE reduction from 1.31 mm to 0.19 mm with feedforward compensation after calibration, implying that post-swap recovery may require both tension re-establishment and compensation-state updates when routing conditions change.

For post-swap tension management specifically, Singh and Khatait [26] provide relevant but indirect evidence via an RCM tension-adjustment mechanism enabling independent cable pretension tuning with invariant routed tendon length. The key limitation is translational scope: this is not a direct quick-change end-effector interface study and does not quantify repeated dock–undock surgical workflow performance.

A comparative summary of the main tool-exchange strategy families discussed in the literature is provided in Table 1.

| Criterion | Rail-guided sliding | Quick-lock / bayonet | Reconfigurable tool |
|---|---|---|---|
| **Mechanism concept** | Linear cartridge-style or rail-guided insertion and removal | Rotational locking, latch-based docking, or compact quick-coupling interface | Single distal tool changes functional mode instead of being physically exchanged |
| **Workflow benefit** | Structured and repeatable alignment during exchange; compatible with trocar / RCM constraints | Potentially compact and rapid attachment with fewer gross insertion steps | Reduces exchange frequency by combining multiple functions into one tool |
| **Main limitation** | Still requires reliable docking, reconnection, and post-swap recovery | Direct evidence for repeatability, sterilisation robustness, and re-zeroing remains limited | Typically validated only for narrow task classes; unlikely to replace the full instrument set |
| **Evidence base** | Strongest direct exchange evidence in the reviewed literature | Sparse direct surgical validation and weak comparative evidence | Promising prototype/task-specific evidence, but limited generality across procedures |
| **Validation maturity** | High | Low | Moderate |

Table 1: Comparison of tool-exchange strategy families relevant to modular tendon-driven surgical manipulators

### 2.3.7 Critical Assessment and Identified Knowledge Gap

The literature supports a clear hierarchy of maturity: (i) strongest evidence for linear, rail-guided exchange, (ii) promising but task-limited multifunctional substitution, and (iii) weak comparative evidence for alternative docking and locking mechanisms.

The unresolved research gap is therefore the integrated validation of *mechanism and control recovery* following repeated tool exchanges. In particular, exchange time, docking repeatability, post-swap backlash and deadband behaviour, retensioning effort, and distal force-safety consistency must be quantified together under clinically realistic routing and sterilisation conditions.

## 2.4 System Safety and Reliability Considerations

### 2.4.1 Safety Constraint Model and Hazard Taxonomy

Safety in macro–micro MIS is treated here as constrained execution under mechanical, control, and workflow limits. In minimally invasive surgery, tools are introduced through a *trocar*, a fixed entry port that enforces a pivoting constraint on the instrument shaft. This is commonly modelled as a remote centre of motion (RCM) constraint, requiring zero velocity at the entry point [12], [13].

At the kinematic level, trocar (RCM) constraint enforcement can be represented by

$$\mathbf{v} * RCM = J * RCM\dot{\mathbf{q}} = \mathbf{0}, \tag{3}$$

and in constrained form

$$\dot{\mathbf{x}}_e = J_c\dot{\mathbf{q}} * I, \quad J_c = J_eT\left[I_n \ - J * II^{-1}J_I\right], \tag{4}$$

which enforces motion consistency under kinematic constraints [13].

For moving trocar conditions, such as soft-tissue interaction or external disturbance, the constraint can be extended to

$$\dot{\mathbf{q}} * II = J * II^{-1}\left(\mathbf{v}_{trc} - J_I\dot{\mathbf{q}}_I\right), \tag{5}$$

preserving explicit constraint compensation under disturbance.

These equations define one hazard layer (constraint violation), but safety risk in this thesis is broader and includes distal force overshoot, device malfunction, post-swap state mismatch, delayed recovery from recoverable faults, and degradation from repeated reprocessing.

### 2.4.2 Error Detection and Reporting Architecture

Rajih, Tholomier, Cormier, *et al.* [27] provide useful evidence on practical malfunction detection, as errors were captured through da Vinci Si event logs, time-stamped, and classified as recoverable versus unrecoverable across 1228 procedures. This is important for this review because safety analysis depends on what is actually detected and recorded, not only what fails physically.

Their workflow separates events by timing (pre-anesthesia setup, intraoperative, and post-undocking) and by subsystem category, creating a useful template for future macro–micro logging: event source, recoverability, corrective action, and impact on operative flow [27].

#### 2.4.3 Malfunction Frequency and Immediate Clinical Consequence

#### 2.4.4 Malfunction Frequency and Clinical Consequence

Across the same cohort, Rajih, Tholomier, Cormier, *et al.* [27] report overall malfunctions in 4.97% (61/1228) of cases, with recoverable errors dominant (46/61). The most common event type was arm pressure/output limit faults (2.04%, 25/1228), followed by communication-related errors (1.06%, 13/1228), failed encoder errors (0.57%, 7/1228), illuminator faults (0.33%, 4/1228), and smaller electronic categories.

In immediate outcome terms, they report no malfunction-driven case abortion, no malfunction-driven conversion, no injury, and no death, with one delayed case due to a battery-related error. This supports their conclusion that malfunctions are not rare but are typically low-consequence when recovery pathways are present. However, this does not eliminate system-level risk, as consequence severity depends on context, detection latency, and available technical support [27].

To maintain explicit risk representation, this review uses

$$\hat{p} * i = \frac{N_i}{N * tot}, \qquad RPN_i = S_i O_i D_i, \tag{6}$$

where $\hat{p}_i$ denotes category occurrence probability and $RPN_i$ is a risk-priority metric based on severity ($S_i$), occurrence ($O_i$), and detectability ($D_i$), with confidence caveats for single-centre and voluntary-reporting datasets.

#### 2.4.5 Preventable Versus Non-Preventable Faults

The Canadian dataset helps separate preventable execution faults from non-preventable electronic anomalies. Rajih, Tholomier, Cormier, *et al.* [27] identify collision and jerky-motion-related events as a major recoverable class and explicitly link their frequency to team learning curve and technique. In contrast, communication-board anomalies are treated as less predictable and often require manufacturer-mediated service.

This distinction is directly relevant for thesis design because mitigation strategies differ: port and arm planning, as well as operator training, can reduce preventable faults, whereas non-preventable faults require rapid diagnosis, robust recoverability logic, maintenance policy, and fallback workflow design [27].

#### 2.4.6 Lifecycle Reliability: Reconfiguration, Wear, and Reprocessing

Short-horizon controller performance remains necessary but insufficient for comprehensive safety claims. Sadeghian, Zokaei, and Hadian Jazi [28] demonstrate feasible constrained tracking performance (including experimental task-space error on the order of $\pm 0.4, \mathrm{mm}$), but this does not address prolonged workflow stress or repeated reconfiguration effects.

For contamination risk, Saito, Yasuhara, Murakoshi, *et al.* [29] show that residual protein levels remain substantially higher in robotic instruments than in conventional instruments after in-house cleaning (robotic: 650, 550, 530 $\mu$g/instrument vs. ordinary: 16, 17, 17 $\mu$g/instrument; $P < .0001$), with an estimated total residual burden of approximately $2.8 \times 10^3, \mu$g versus 98 $\mu$g. The practical implication is that sterilisation and reprocessing must be treated as tracked reliability variables when repeated tool exchange and reuse are core to platform operation.

### 2.4.7 Integrated Safety Gap and Evaluation Endpoints

The supported safety gap is not the absence of individual solutions; rather, it is the lack of integrated validation across *constraint control, malfunction detection and recovery, and lifecycle reprocessing reliability* within a single coherent macro–micro workflow.

Accordingly, later chapters should report unified evaluation endpoints rather than isolated metrics. These include trocar-point (RCM) error bounds, malfunction detection-to-recovery time, post-swap recovery time, residual deadband and backlash after reconfiguration, distal-force limit adherence, repeatability across repeated cycles, and post-reprocessing contamination bounds.

## 2.5 Synthesis of Literature and Knowledge Gaps

The reviewed literature establishes a coherent but incomplete technical chain for macro–micro MIS systems. Continuum and tendon-driven hardware studies support distal dexterity in constrained anatomy, teleoperation and control studies show that tremor, delay, estimation uncertainty, and intent mapping are jointly performance-critical, and safety studies demonstrate that malfunction events are generally manageable but not negligible. Accessibility studies then add an implementation constraint: for translation, performance must be achieved under realistic cost, maintenance, and workforce limits, not only in high-resource laboratory settings.

Critically, these bodies of work are usually validated in isolation. Kinematic and compensation studies often assume controlled routing and loading conditions; teleoperation studies typically assume simplified communication conditions or bounded tasks; malfunction studies are strong in event reporting but weaker in mechanism-level causal interpretation; and lifecycle reprocessing studies quantify residual burden but are rarely connected to control-state recovery following repeated instrument exchange. This fragmentation limits the explanatory strength of any single study when the research question concerns end-to-end workflow reliability.

In relation to this project specifically, the conceptual relationship is direct. The UR5e macro stage provides accessible gross positioning, while the continuum distal stage provides local dexterity; however, the interface between them introduces coupled failure modes, including vibration transmission, backlash and hysteresis, post-swap recalibration burden, and safety assurance under disturbance. The research problem is therefore not limited to improving a single controller or mechanism, but to determining whether a low-cost macro–micro architecture can preserve safety and repeatability when control, mechanics, and reconfiguration are exercised together.

Based on this synthesis, the principal knowledge gaps are:

- **Integrated validation gap:** limited evidence combining constrained motion control, teleoperation robustness, and modular reconfiguration recovery within a single experimental framework.
- **Safety evidence gap:** limited prospective linkage between detected robotic malfunctions, recovery actions, and task-level outcome degradation under realistic operating conditions.
- **Lifecycle reliability gap:** insufficient long-horizon data on wear, sterilisation and reprocessing residue, and calibration drift across repeated swap cycles for tendon-driven instruments.
- **Benchmarking gap:** insufficient standardised and reproducible metrics and protocols enabling fair comparison of low-cost macro–micro platforms using clinically meaningful endpoints.

In this context, the thesis contribution is positioned as a systems-level evaluation: to design and assess a UR5e-based macro–micro platform using unified endpoints for constraint adherence, teleoperation robustness, post-swap recovery, and lifecycle reliability, rather than reporting isolated improvements in individual subsystems.

### 2.5.1 Remaining Evidence Gaps and Future Directions

The literature can be further strengthened by recent work in several areas:

- benchmark datasets and standardised test protocols for continuum surgical teleoperation,
- variable-delay and packet-loss-robust bilateral and shared control in human-in-the-loop settings,
- safety assurance frameworks (fault handling, bounded motion, and force-limited behaviour) for MIS robots,
- quantitative studies of modular tool-exchange repeatability and calibration drift,
- lifecycle reliability of tendon-driven instruments under repeated use and sterilisation.

# 3 Research Question and Project Plan

## 3.1 Research Question and Plan Status

This section is maintained as a separate planning section that draws on the literature review but is not part of the literature-review narrative itself. At this stage, the research question and plan are presented with sufficient detail to indicate feasibility, expected resource requirements, and realistic time demands.

## 3.2 Plan Quality Criteria (Assessment-Aligned)

The planning section is intended to demonstrate the following progression:

- The research question and plan are clearly presented with enough detail to support feasibility and basic thesis structure.
- The plan is explicitly informed by the reviewed literature, with a clear explanation of why each major work package follows from identified evidence gaps.
- The plan includes realistic milestones, a logical thesis outline, and direct links between chapter flow, reviewed evidence, and planned validation.
- The plan includes contingencies for project variation (e.g., hardware delays, integration issues, data-quality limitations) and alternative execution pathways.

## 3.3 Current Planning Position

The current project plan is retained as a standalone section for subsequent development. The immediate objective is to preserve alignment with the literature-derived gaps while deferring detailed milestone refinement to the next planning pass.

# 4 Project Dependent Preparations

All required project-dependent preparations should be tracked here and reviewed for readiness before execution. This section should confirm whether preparations are complete and whether significant revision is still required. It should also record exceptional early progress where available, so that planning assumptions can be confirmed or adjusted and preliminary answers to research questions can be documented.

# 5 Conclusion

The reviewed literature shows a clear convergence on three points. First, macro-micro architectures and tendon-driven continuum tools are well motivated for MIS dexterity, but remain sensitive to vibration transmission, modelling error, and teleoperation instability. Second, modern control strategies (filtering, latency compensation, predictive bilateral control, and intent-aware shared autonomy) improve performance, yet are often validated under constrained assumptions that limit direct transfer to clinically variable settings. Third, cost and adoption studies indicate that accessibility is governed by full-system burden (maintenance, disposables, training continuity, and infrastructure), not capital cost alone.

Taken together, the strongest unresolved knowledge gaps are integrated, reproducible frameworks that jointly address delay-robust control, modular reconfiguration repeatability, safety-critical reliability, and realistic cost-constrained deployment conditions.

# Acknowledgements

*I would like to thank...*

# References

# References

[1] L. Scott, "Design and development of a macro-micro robotic system for minimally invasive surgery," unpublished, provided by prof. Liao Wu, Honours thesis, UNSW Sydney, 2023.

[2] V. Vitiello, S.-L. Lee, T. P. Cundy, and G.-Z. Yang, "Emerging robotic platforms for minimally invasive surgery," *IEEE Reviews in Biomedical Engineering*, vol. 6, pp. 111–126, 2013. DOI: `10.1109/RBME.2012.2236311`.

[3] R. Hargest, "Five thousand years of minimal access surgery: 1850 to 1990: Technological developments," *Journal of the Royal Society of Medicine*, vol. 114, no. 1, pp. 19–29, 2021. DOI: `10.1177/0141076820967918`.

[4] M. Hatzinger, A. Fesenko, and M. Sohn, "The first human laparoscopy and NOTES operation: Dimitrij oscarovic ott (1855–1929)," *Urologia Internationalis*, vol. 92, no. 4, pp. 387–391, 2014. DOI: `10.1159/000358016`.

[5] Intuitive, *Da vinci robotic surgical systems*. [Online]. Available: `https://www.intuitive.com/en-us/products-and-services/da-vinci` (visited on 04/19/2026).

[6] D. Ko, Y. Kim, H. Lim, and S. Kim, "Design of automatic tool replacement mechanism for laparoscopic surgical robot arm for solo surgery," *The International Journal of Medical Robotics and Computer Assisted Surgery*, vol. 21, no. 5, e70106, 2025. DOI: `10.1002/rcs.70106`.

[7] J. Seetohul and M. Shafiee, "Snake robots for surgical applications: A review," *Robotics*, vol. 11, no. 3, p. 57, 2022. DOI: `10.3390/robotics11030057`.

[8] A. Razjigaev, A. K. Pandey, D. Howard, J. Roberts, and L. Wu, "SnakeRaven: Teleoperation of a 3d printed snake-like manipulator integrated to the RAVEN II surgical robot," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 5282–5288. DOI: `10.1109/IROS51168.2021.9636878`.

[9] J. A. Eckhoff, D. T. Mueller, S. N. Brunner, H. F. Fuchs, and O. R. Meireles, "Do the costs of robotic surgery present an insurmountable obstacle? a narrative review," *International Journal of Abdominal Wall and Hernia Surgery*, vol. 6, no. 2, p. 71, 2023. DOI: `10.4103/ijawhs.IJAWHS_62_22`.

[10] Z. Raza, S. U. Saqib, H. Zafar, and S. Islam, "Transforming surgical landscapes: Obstacles in the integration of robotic surgery in pakistan – is cost the only limiting factor?" *Journal of the Pakistan Medical Association*, vol. 74, no. Suppl. 4, S151–S157, 2024. DOI: `10.47391/JPMA.AKU-9S-23`.

[11] Standard Bots, *Universal robots price guide: What to expect (new and used costs).* [Online]. Available: https://standardbots.com/blog/universal-robot-price (visited on 04/20/2026).

[12] J. Burgner-Kahrs, D. C. Rucker, and H. Choset, "Continuum robots for medical applications: A survey," *IEEE Transactions on Robotics*, vol. 31, no. 6, pp. 1261–1280, 2015. DOI: 10.1109/TRO.2015.2489500.

[13] P. E. Dupont, N. Simaan, H. Choset, and C. Rucker, "Continuum robots for medical interventions," *Proceedings of the IEEE*, vol. 110, no. 7, pp. 847–870, 2022. DOI: 10.1109/JPROC.2022.3141338.

[14] F. Wang, H. Wang, J. Luo, *et al.*, "FIORA: A flexible tendon-driven continuum manipulator for laparoscopic surgery," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 1166–1173, 2022. DOI: 10.1109/LRA.2021.3134276.

[15] H. Sang, C. Yang, F. Liu, J. Yun, G. Jin, and F. Chen, "A zero phase adaptive fuzzy kalman filter for physiological tremor suppression in robotically assisted minimally invasive surgery," *The International Journal of Medical Robotics and Computer Assisted Surgery*, vol. 12, no. 4, pp. 658–669, 2016. DOI: 10.1002/rcs.1741.

[16] T. Batty, A. Ehrampoosh, B. Shirinzadeh, Y. Zhong, and J. Smith, "A transparent teleoperated robotic surgical system with predictive haptic feedback and force modelling," *Sensors*, vol. 22, no. 24, p. 9770, 2022. DOI: 10.3390/s22249770.

[17] M. H. Lashari, W. Batayneh, and A. Khokhar, "Enhancing precision in tactile internet-enabled remote robotic surgery: Kalman filter approach," *arXiv*, 2024. DOI: 10.48550/ARXIV.2406.04503.

[18] M. Ebrahimian, M. Pourmokhtari, M. Ghiyasi, B. Yazdankhoo, and B. Beigzadeh, "Online bilateral predictive control for time-delayed teleoperation of snake-like robots," *Journal of Intelligent & Robotic Systems*, vol. 110, p. 80, 2024. DOI: 10.1007/s10846-024-02113-3.

[19] M. Bowman, J. Zhang, and X. Zhang, "Intent-based task-oriented shared control for intuitive telemanipulation," *Journal of Intelligent & Robotic Systems*, vol. 110, p. 167, 2024. DOI: 10.1007/s10846-024-02185-1.

[20] R. Bogue, "Developments in robotic teleoperation," *Industrial Robot*, vol. 51, no. 5, pp. 697–703, 2024. DOI: 10.1108/IR-08-2024-0358.

[21] S. Zou, G. Li, and Y. Huang, "Transmission backlash compensation and grasping force estimation of surgical instruments for the laparoscopic minimally invasive surgery robot," *Applied Sciences*, vol. 12, no. 23, p. 12 126, 2022. DOI: 10.3390/app122312126.

[22] J. Hong, D. Hong, C. Kim, and S. H. Won, "Modeling backlash-like hysteresis and feedforward control of tendon-sheath mechanism pair with arbitrary shape: Excluding distal sensory input," *Journal of Mechanisms and Robotics*, vol. 17, no. 5, p. 051 009, 2025. DOI: 10.1115/1.4067020.

[23] B. Cowan, C. Gomes, P. Morris, *et al.*, "Robotic technology in surgery: A classification system of soft tissue surgical robotic devices," *Surgical Endoscopy*, vol. 38, no. 7, pp. 3645–3653, 2024. DOI: 10.1007/s00464-024-10861-4.

[24] D. Torielli, L. Bertoni, F. Fusaro, N. Tsagarakis, and L. Muratore, "ROS End-Effector: A hardware-agnostic software and control framework for robotic end-effectors," *Journal of Intelligent & Robotic Systems*, vol. 108, p. 70, 2023. DOI: 10.1007/s10846-023-01911-5.

[25] L. Mocellin, N. Pagliarani, G. Gamberini, G. Ciuti, M. Cianchetti, and A. Menciassi, "OriGrasp: A multifunctional origami-inspired instrument for delicate manipulation in abdominal surgery," *Advanced Intelligent Systems*, e202501249, 2026. DOI: 10.1002/aisy.202501249.

[26] A. Singh and J. P. Khatait, "Mechanism for tension adjustment in tendon driven remote center-of-motion mechanism for minimally invasive surgery," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, vol. 236, no. 16, pp. 9293–9305, 2022. DOI: 10.1177/09544062221093899.

[27] E. Rajih, C. Tholomier, B. Cormier, *et al.*, "Error reporting from the da vinci surgical system in robotic surgery: A canadian multispecialty experience at a single academic centre," *Canadian Urological Association Journal*, vol. 11, no. 5, E197–E202, 2017. DOI: 10.5489/cuaj.4116.

[28] H. Sadeghian, F. Zokaei, and S. Hadian Jazi, "Constrained kinematic control in minimally invasive robotic surgery subject to remote center of motion constraint," *Journal of Intelligent & Robotic Systems*, vol. 95, no. 3–4, pp. 901–913, 2019. DOI: 10.1007/s10846-018-0927-0.

[29] Y. Saito, H. Yasuhara, S. Murakoshi, T. Komatsu, K. Fukatsu, and Y. Uetera, "Challenging residual contamination of instruments for robotic surgery in japan," *Infection Control & Hospital Epidemiology*, vol. 38, no. 2, pp. 143–146, 2017. DOI: 10.1017/ice.2016.249.

# A Rubric-Aligned Coverage Notes (MMAN4951/MMAN945 Interim Report)

### Criterion 1: Literature Review (High Weighting)

To target Distinction/High Distinction bands, this review must do more than summarize papers. It should:

- Show conceptual links between manipulator design, control, safety, and modularity.
- Explicitly identify and justify knowledge gaps your thesis addresses.
- Include recent work (not only foundational papers).
- Critically compare methods instead of listing them.

### Criterion 4: Document Presentation

For marks protection, maintain:

- consistent citation formatting,
- clear figure/table captions,
- polished grammar and structured argument flow.